Tests de l'application

Ingénieur applicatif : Lareine TEGA | Date : 24/10/2026 | GROUPE (.env) : 5, réseau 192.168.60.0/24

Projet RÉSULTATS-BF, Master 1 Informatique, IBAM. Les sorties ci-dessous sont celles relevées sur le poste de l'auteur (captures d'écran du 24/10/2026), recopiées sans retouche. **Statut :** T1, T2, T3, T6 et T7 réalisés ; T5 à compléter (voir son commentaire) ; T4 à faire.

Démarrage de la plateforme

```
    lb1: Warning: Connection reset. Retrying...
    lb1:
    lb1: Vagrant insecure key detected. Vagrant will automatically replace
    lb1: this with a newly generated keypair for better security.
    lb1:
    lb1: Inserting generated public key within guest...
    lb1: Removing insecure key from the guest if it's present...
    lb1: Key inserted! Disconnecting and reconnecting using new SSH key...
==> lb1: Machine booted and ready!
==> lb1: Checking for guest additions in VM...
    lb1: The guest additions on this VM do not match the installed version of
    lb1: VirtualBox! In most cases this is fine, but in rare cases it can
    lb1: prevent things such as shared folders from working properly. If you see
    lb1: shared folder errors, please make sure the guest additions within the
    lb1: virtual machine match the version of VirtualBox you have installed on
    lb1: your host and reload your VM.
    lb1:
    lb1: Guest Additions Version: 7.0.16
    lb1: VirtualBox Version: 7.2
==> lb1: Setting hostname...
==> lb1: Configuring and enabling network interfaces...
==> lb1: Mounting shared folders...
    lb1: C:/Users/Lareine_service/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf => /vagrant
==> lb1: Running provisioner: shell...
    lb1: Running: C:/Users/LAREIN~1/AppData/Local/Temp/vagrant-shell20261024-42676-rkocuh.sh
==> lb1: Running provisioner: shell...
    lb1: Running: C:/Users/LAREIN~1/AppData/Local/Temp/vagrant-shell20261024-42676-e5mqkh.sh
    lb1: haproxy.cfg mis à jour

C:\Users\Lareine_service\Downloads\M1SIE\TP_VIRTUALISATION\resultats-bf>
```

Fin de vagrant up : les cinq machines sont provisionnées, lb1 a généré haproxy.cfg.

```
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (main)
$ curl http://localhost:8080/info
curl "http://localhost:8080/resultat?matricule=BF000123"
{"hote": "app1"}{"matricule": "BF000123", "nom": "Candidat 123", "serie": "E", "moyenne": 9.05, "admis": false}
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (main)
$ curl http://localhost:8080/info
{"hote": "app1"}
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (main)
$
```

Vérifications initiales via le répartiteur (/info et /resultat).

T1. Matricule manquant

Commande

```
curl -i "http://localhost:8080/resultat"
```

Attendu

400 et {"erreur": "matricule manquant"}

Sortie brute

```
HTTP/1.0 400 Bad Request
server: BaseHTTP/0.6 Python/3.12.3
date: Sat, 24 Oct 2026 11:04:43 GMT
content-type: application/json; charset=utf-8
content-length: 32
connection: keep-alive

{"erreur": "matricule manquant"}
```

Commentaire

Conforme. L'application refuse proprement une requête incomplète, avec un JSON explicite.

T2. Matricule inconnu

Commande

```
curl -i "http://localhost:8080/resultat?matricule=XX"
```

Attendu

404 et {"erreur": "matricule inconnu"}

Sortie brute

```
HTTP/1.0 404 Not Found
server: BaseHTTP/0.6 Python/3.12.3
date: Sat, 24 Oct 2026 11:04:45 GMT
content-type: application/json; charset=utf-8
content-length: 31
connection: keep-alive

{"erreur": "matricule inconnu"}
```

Commentaire

Conforme. La base répond mais aucune ligne ne correspond : l'application distingue ce cas d'une panne (503).

T3. Matricule valide

Commande

```
curl -i "http://localhost:8080/resultat?matricule=BF000123"
```

Attendu

200 et un JSON avec matricule, nom, serie, moyenne, admis

Sortie brute

```
HTTP/1.0 200 OK
server: BaseHTTP/0.6 Python/3.12.3
date: Sat, 24 Oct 2026 11:04:45 GMT
content-type: application/json; charset=utf-8
content-length: 95
connection: keep-alive

{"matricule": "BF000123", "nom": "Candidat 123", "serie": "E", "moyenne": 9.05, "admis": false}
```

Commentaire

Conforme. Les cinq champs de la table resultats sont présents et admis est cohérent avec la moyenne (9,05, inférieure à 10).



Capture commune à T1, T2 et T3.

T4. Route inconnue

Commande

```
curl -i http://localhost:8080/nimporte
```

Attendu

404 et {"erreur": "route inconnue"}

Sortie brute

À faire

Commentaire

À faire.

T5. Absence d'état : redémarrage d'un serveur applicatif

Commande

```
vagrant ssh app1 -c "sudo systemctl restart resultats-bf"
curl -i "http://localhost:8080/resultat?matricule=BF000123"
```

Attendu

La réponse revient immédiatement, identique à T3, sans autre action.

Sortie brute

```
HTTP/1.0 200 OK
server: BaseHTTP/0.6 Python/3.12.3
date: Sat, 24 Oct 2026 11:08:42 GMT
content-type: application/json; charset=utf-8
content-length: 95
connection: keep-alive

{"matricule": "BF000123", "nom": "Candidat 123", "serie": "E", "moyenne": 9.05, "admis": false}
```

Commentaire

La réponse est identique à T3. Limite de ce test : la requête passe par le répartiteur, qui peut l'avoir confiée à app2 plutôt qu'à app1. Pour prouver qu'app1 répond aussitôt après redémarrage, refaire le test directement sur http://192.168.60.21:3000/resultat?matricule=BF000123 (sortie à ajouter).

```
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (main)
$ curl http://localhost:8080/info
curl "http://localhost:8080/resultat?matricule=BF000123"
{"hote": "app1"}{"matricule": "BF000123", "nom": "Candidat 123", "serie": "E", "moyenne": 9.05, "admis": false}
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (main)
$ curl http://localhost:8080/info
{"hote": "app1"}
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (main)
$ curl -i "http://localhost:8080/resultat"
curl -i "http://localhost:8080/resultat?matricule=XX"
curl -i "http://localhost:8080/resultat?matricule=BF000123"
HTTP/1.0 400 Bad Request
server: BaseHTTP/0.6 Python/3.12.3
date: Sat, 24 Oct 2026 11:04:43 GMT
content-type: application/json; charset=utf-8
content-length: 32
connection: keep-alive

{"erreur": "matricule manquant"}HTTP/1.0 404 Not Found
server: BaseHTTP/0.6 Python/3.12.3
date: Sat, 24 Oct 2026 11:04:45 GMT
content-type: application/json; charset=utf-8
content-length: 31
connection: keep-alive

{"erreur": "matricule inconnu"}HTTP/1.0 200 OK
server: BaseHTTP/0.6 Python/3.12.3
date: Sat, 24 Oct 2026 11:04:45 GMT
content-type: application/json; charset=utf-8
content-length: 95
connection: keep-alive

{"matricule": "BF000123", "nom": "Candidat 123", "serie": "E", "moyenne": 9.05, "admis": false}
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$ vagrant ssh app1 -c "sudo systemctl restart resultats-bf"
curl -i "http://localhost:8080/resultat?matricule=BF000123"
HTTP/1.0 200 OK
server: BaseHTTP/0.6 Python/3.12.3
date: Sat, 24 Oct 2026 11:08:42 GMT
content-type: application/json; charset=utf-8
content-length: 95
connection: keep-alive

{"matricule": "BF000123", "nom": "Candidat 123", "serie": "E", "moyenne": 9.05, "admis": false}
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$
```

T5 : redémarrage d'app1 puis lecture via le répartiteur.

T6. Base absente : /sante renvoie 503 proprement

Commande

```
vagrant halt bdd1
sleep 5
curl -i http://192.168.60.21:3000/sante
curl -i http://192.168.60.21:3000/info
vagrant up bdd1
```

Attendu

/sante en 503 avec un JSON, jamais une page blanche ; /info en 200 (cette route ne touche pas la base).

Sortie brute

```
$ vagrant halt bdd1
==> bdd1: Attempting graceful shutdown of VM...
==> bdd1: Forcing shutdown of VM...
$ sleep 5
$ curl -i http://192.168.60.21:3000/sante
HTTP/1.0 503 Service Unavailable
Server: BaseHTTP/0.6 Python/3.12.3
Date: Sat, 24 Oct 2026 11:20:34 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 31

{"erreur": "base indisponible"}
$ curl -i http://192.168.60.21:3000/info
HTTP/1.0 200 OK
```

```
Server: BaseHTTP/0.6 Python/3.12.3
Date: Sat, 24 Oct 2026 11:20:56 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 16

{"hote": "app1"}
```

Commentaire

Conforme. Quand la base primaire est arrêtée, l'application répond 503 avec un message JSON, et /info reste à 200. C'est la sonde que lit le répartiteur : un 503 fait retirer le serveur du service. Incident de manipulation sans lien avec l'application : une première exécution a utilisé l'adresse littérale 192.168.X.21 (Could not resolve host), car le X n'avait pas été remplacé par 60 ; le test a été refait avec la bonne adresse. Temps de réponse de /sante : à mesurer.

```
curl: (6) Could not resolve host: 192.168.X.21
Bringing machine 'bdd1' up with 'virtualbox' provider...
==> bdd1: Clearing any previously set forwarded ports...
==> bdd1: Clearing any previously set network interfaces...
==> bdd1: Preparing network interfaces based on configuration...
    bdd1: Adapter 1: nat
    bdd1: Adapter 2: hostonly
==> bdd1: Forwarding ports...
    bdd1: 22 (guest) => 2222 (host) (adapter 1)
==> bdd1: Running 'pre-boot' VM customizations...
==> bdd1: Booting VM...
==> bdd1: Waiting for machine to boot. This may take a few minutes...
    bdd1: SSH address: 127.0.0.1:2222
    bdd1: SSH username: vagrant
    bdd1: SSH auth method: private key
==> bdd1: Machine booted and ready!
==> bdd1: Checking for guest additions in VM...
    bdd1: The guest additions on this VM do not match the installed version of
    bdd1: VirtualBox! In most cases this is fine, but in rare cases it can
    bdd1: prevent things such as shared folders from working properly. If you see
    bdd1: shared folder errors, please make sure the guest additions within the
    bdd1: virtual machine match the version of VirtualBox you have installed on
    bdd1: your host and reload your VM.
    bdd1:
    bdd1: Guest Additions Version: 7.0.16
    bdd1: VirtualBox Version: 7.2
==> bdd1: Setting hostname...
==> bdd1: Configuring and enabling network interfaces...
==> bdd1: Mounting shared folders...
    bdd1: C:/Users/Lareine_service/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf => /vagrant
==> bdd1: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> bdd1: flag to force provisioning. Provisioners marked to run always will still run.

Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$ vagrant halt bdd1
==> bdd1: Attempting graceful shutdown of VM...
==> bdd1: Forcing shutdown of VM...

Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$ sleep 5

Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$ curl -i http://192.168.X.21:3000/sante
curl: (6) Could not resolve host: 192.168.X.21

Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$ curl -i http://192.168.60.21:3000/sante
HTTP/1.0 503 Service Unavailable
Server: BaseHTTP/0.6 Python/3.12.3
Date: Sat, 24 Oct 2026 11:20:34 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 31

{"erreur": "base indisponible"}
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$ curl -i http://192.168.60.21:3000/info
HTTP/1.0 200 OK
Server: BaseHTTP/0.6 Python/3.12.3
Date: Sat, 24 Oct 2026 11:20:56 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 16

{"hote": "app1"}
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$
```

T6 : base arrêtée, 503 sur /sante, 200 sur /info.

T7. Aucun secret dans l'historique Git (R7)

Commande

```
git log -p | grep -i password
```

Attendu

Uniquement des noms de variables ou la valeur factice changez_moi.

Sortie brute

```
+DB_PASSWORD=changez_moi
+REPL_PASSWORD=changez_moi_aussi
+GRAFANA_ADMIN_PASSWORD=changez_moi_encore
+  "DB_PASSWORD" => cfg("DB_PASSWORD", ""),
+  "REPL_PASSWORD" => cfg("REPL_PASSWORD", ""),
+  "GRAFANA_ADMIN_PASSWORD" => cfg("GRAFANA_ADMIN_PASSWORD", "admin"),
+        user=os.environ["DB_USER"], password=os.environ["DB_PASSWORD"],
+- Vérifier l'historique : `git log --oneline` (4 auteurs) et `git log -p | grep -i password` ne
    doit montrer aucun vrai mot de passe.
+DB_PASSWORD=${DB_PASSWORD}
+psql_pg -c "ALTER ROLE ${DB_USER} PASSWORD '${DB_PASSWORD}'"
+psql_pg -c "ALTER ROLE ${REPL_USER} PASSWORD '${REPL_PASSWORD}'"
+echo "GF_SECURITY_ADMIN_PASSWORD=${GRAFANA_ADMIN_PASSWORD}" > /etc/default/grafana-server
+    -d "host=${RESEAU}.31 user=${REPL_USER} password=${REPL_PASSWORD}" \
```

Commentaire

Aucun vrai mot de passe. Les mots de passe passent par des variables (${DB_PASSWORD}, os.environ[...]) et .env.exemple ne contient que des valeurs factices. À noter : le Vagrantfile utilise "admin" comme mot de passe Grafana par défaut si .env n'en définit pas ; ce n'est pas un secret réel, mais c'est un mot de passe faible par défaut. Vérification complémentaire à faire : git ls-files | grep -x .env ne doit rien afficher.

```
Content-Type: application/json; charset=utf-8
Content-Length: 31

{"erreur": "base indisponible"}
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$ curl -i http://192.168.60.21:3000/info
HTTP/1.0 200 OK
Server: BaseHTTP/0.6 Python/3.12.3
Date: Sat, 24 Oct 2026 11:20:56 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 16

{"hote": "app1"}
Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$ vagrant up bdd1
Bringing machine 'bdd1' up with 'virtualbox' provider...
==> bdd1: Clearing any previously set forwarded ports...
==> bdd1: Clearing any previously set network interfaces...
==> bdd1: Preparing network interfaces based on configuration...
    bdd1: Adapter 1: nat
    bdd1: Adapter 2: hostonly
==> bdd1: Forwarding ports...
    bdd1: 22 (guest) => 2222 (host) (adapter 1)
==> bdd1: Running 'pre-boot' VM customizations...
==> bdd1: Booting VM...
==> bdd1: Waiting for machine to boot. This may take a few minutes...
    bdd1: SSH address: 127.0.0.1:2222
    bdd1: SSH username: vagrant
    bdd1: SSH auth method: private key
==> bdd1: Machine booted and ready!
==> bdd1: Checking for guest additions in VM...
    bdd1: The guest additions on this VM do not match the installed version of
    bdd1: VirtualBox! In most cases this is fine, but in rare cases it can
    bdd1: prevent things such as shared folders from working properly. If you see
    bdd1: shared folder errors, please make sure the guest additions within the
    bdd1: virtual machine match the version of VirtualBox you have installed on
    bdd1: your host and reload your VM.
    bdd1:
    bdd1: Guest Additions Version: 7.0.16
    bdd1: VirtualBox Version: 7.2
==> bdd1: Setting hostname...
==> bdd1: Configuring and enabling network interfaces...
==> bdd1: Mounting shared folders...
    bdd1: C:/Users/Lareine_service/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf => /vagrant
==> bdd1: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> bdd1: flag to force provisioning. Provisioners marked to run always will still run.

Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$ git log -p | grep -i password
+DB_PASSWORD=changez_moi
+REPL_PASSWORD=changez_moi_aussi
+GRAFANA_ADMIN_PASSWORD=changez_moi_encore
+  "DB_PASSWORD" => cfg("DB_PASSWORD", ""),
+  "REPL_PASSWORD" => cfg("REPL_PASSWORD", ""),
+  "GRAFANA_ADMIN_PASSWORD" => cfg("GRAFANA_ADMIN_PASSWORD", "admin"),
+        user=os.environ["DB_USER"], password=os.environ["DB_PASSWORD"],
+- Vérifier l'historique : `git log --oneline` (4 auteurs) et `git log -p | grep -i password` ne doit montrer aucun vrai mot de passe.
+DB_PASSWORD=${DB_PASSWORD}
+psql_pg -c "ALTER ROLE ${DB_USER} PASSWORD '${DB_PASSWORD}'"
+psql_pg -c "ALTER ROLE ${REPL_USER} PASSWORD '${REPL_PASSWORD}'"
+echo "GF_SECURITY_ADMIN_PASSWORD=${GRAFANA_ADMIN_PASSWORD}" > /etc/default/grafana-server
+    -d "host=${RESEAU}.31 user=${REPL_USER} password=${REPL_PASSWORD}" \

Lareine_service@DESKTOP-1PQ5LOA MINGW64 ~/Downloads/M1SIE/TP_VIRTUALISATION/resultats-bf (feat/tests-app)
$
```

T7 : historique Git, aucun secret réel.

Anomalies constatées et propositions

Test	Constat	Proposition
T3	La moyenne de BF000123 vaut 9,05 sur ce poste, contre 19,73 sur le poste d'Adama.	Les 50 000 lignes sont engendrées avec random() : les données changent à chaque reconstruction. Pour des mesures reproductibles, fixer la graine (SELECT setseed(0.42); avant l'INSERT de scripts/bdd.sh).
T5	Le test passe par le répartiteur et ne prouve pas qu'app1 répond après redémarrage.	Refaire le test directement sur 192.168.60.21:3000.
T7	Mot de passe Grafana par défaut "admin" dans le Vagrantfile.	Exiger la variable dans .env ou supprimer la valeur par défaut.

Conclusion

Les trois codes d'erreur de l'application (400, 404, 503) sont conformes à l'attendu et la réponse JSON est toujours propre. Le serveur ne garde aucun état local, et la base absente est signalée par /sante sans que /info en souffre. Restent à compléter : T4, le test direct de T5 et le temps de réponse de T6.